



UNIVERSITÀ DEGLI STUDI DI PISA

CORSO DI STUDIO IN BIONICS ENGINEERING

STATISTICAL SIGNAL PROCESSING - *Prova scritta del 20/07/2023*

EXERCISE

It is given a w.s.s. zero mean Moving Average (MA) process $S[n]$ of order 1, $S[n] = I[n] - \alpha I[n-1]$, where $I[n]$ is the zero-mean innovation process with power σ_I^2 .

(1) Derive the system function $L(z)$ of the innovation filter, its zeros and poles, as a function of α ; determine if when $\alpha = 0.8$ it is a low-pass, high-pass, or band-pass filter. Derive the autocorrelation function (ACF) $R_S[m] = E\{S[n]S[n+m]\}$ and the power spectral density (PSD) $S_S(e^{j2\pi f}) = FT\{R_S[m]\}$ of the process $S[n]$ as a function of α and plot them for $\alpha = 0.8$ and $\sigma_I^2 = 2$.

(2) Apply the orthogonality principle to derive the linear minimum mean square error (LMMSE) estimator of $S[n]$ given the observed data $S[n-1]$ and $S[n-2]$ and its mean square error (MSE) $\sigma_\epsilon^2 \triangleq E\{\epsilon^2[n]\} = E\{(S[n] - \hat{S}[n])^2\}$ as a function of α . Calculate them for $\alpha = 0.8$ and $\sigma_I^2 = 2$.

(3) Now assume that $\alpha = 0.8$, $\sigma_I^2 = 2$, and that the signal of interest $S[n]$ is observed in the presence of additive white noise, i.e. $X[n] = S[n] + W[n]$, where $W[n]$ is additive white noise with power $\sigma_W^2 = 1$. Apply the orthogonality principle to derive the LMMSE estimator of $S[n]$ given the noisy observed data $X[n-1]$ and $X[n-2]$; derive its MSE and compare it with the one previously derived.



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Solution.

$$(1) S[n] = I[n] - \alpha I[n-1]$$

$$S(z) = I(z) - \alpha z^{-1} I(z)$$

$$L(z) = \frac{S(z)}{I(z)} = 1 - \alpha z^{-1} = \frac{z - \alpha}{z}$$

The innovation filter has 1 zero at $z = \alpha$ and 1 pole at $z = 0$. Hence, when $\alpha = 0.8$ the innovation filter is a high-pass filter, and so the process is a high-pass process.

The ACF and the PSD are given by:

$$\begin{aligned} R_s[m] &= E\{S[n]S[n+m]\} = E\{(I[n] - \alpha I[n-1])(I[n+m] - \alpha I[n+m-1])\} \\ &= R_I[m] - \alpha R_I[m-1] - \alpha R_I[m+1] + \alpha^2 R_I[m] = (1 + \alpha^2) \sigma_I^2 \delta[m] - \alpha \sigma_I^2 \delta[m-1] - \alpha \sigma_I^2 \delta[m+1] \\ &\stackrel{\alpha=0.8}{=} 3.28 \cdot \delta[m] - 1.6 \cdot \delta[m-1] - 1.6 \cdot \delta[m+1] \end{aligned}$$

$$R_s[m] = \begin{cases} (1 + \alpha^2) \sigma_I^2, & m = 0 \\ -\alpha \sigma_I^2, & m = \pm 1 \\ 0, & |m| \geq 2 \end{cases} \stackrel{\alpha=0.8}{=} \begin{cases} 3.28, & m = 0 \\ -1.6, & m = \pm 1 \\ 0, & |m| \geq 2 \end{cases}$$

$$\begin{aligned} S_s(e^{j2\pi f}) &= FT\{R_s[m]\} = \sum_{m=-\infty}^{+\infty} R_s[m] e^{-j2\pi f m} \\ &= \sum_{m=-\infty}^{+\infty} [(1 + \alpha^2) \sigma_I^2 \delta[m] - \alpha \sigma_I^2 \delta[m-1] - \alpha \sigma_I^2 \delta[m+1]] e^{-j2\pi f m} \\ &= \sigma_I^2 (1 + \alpha^2 - \alpha e^{-j2\pi f} - \alpha e^{j2\pi f}) = \sigma_I^2 (1 + \alpha^2 - 2\alpha \cos(2\pi f)) \\ &\stackrel{\alpha=0.8}{=} 3.28 - 3.2 \cos(2\pi f) \end{aligned}$$

When $\alpha = 0.8$, $S[n]$ is a high-pass process.

$$(2) \hat{S}[n] = h[1]S[n-1] + h[2]S[n-2] + b$$

Since all the processes are zero mean we have: $E\{\varepsilon[n]\} = 0 \Rightarrow b = 0$.



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$$E\{\varepsilon[n]S[n-m]\} = 0, \quad m=1,2$$

$$E\{(S[n] - \hat{S}[n])S[n-m]\} = 0, \quad m=1,2$$

$$E\{(S[n] - h[1]S[n-1] - h[2]S[n-2])S[n-m]\} = 0, \quad m=1,2$$

$$R_s[m] - h[1]R_s[m-1] - h[2]R_s[m-2] = 0, \quad m=1,2$$

$$\begin{cases} h[1]R_s[0] + h[2]R_s[-1] = R_s[1] \\ h[1]R_s[1] + h[2]R_s[0] = R_s[2] \end{cases} \rightarrow \begin{bmatrix} R_s[0] & R_s[-1] \\ R_s[1] & R_s[0] \end{bmatrix} \begin{bmatrix} h[1] \\ h[2] \end{bmatrix} = \begin{bmatrix} R_s[1] \\ R_s[2] \end{bmatrix}$$

$$\begin{bmatrix} 1+\alpha^2 & -\alpha \\ -\alpha & 1+\alpha^2 \end{bmatrix} \begin{bmatrix} h[1] \\ h[2] \end{bmatrix} = \begin{bmatrix} -\alpha \\ 0 \end{bmatrix} \rightarrow \begin{bmatrix} h[1] \\ h[2] \end{bmatrix} = \frac{1}{(1+\alpha^2)^2 - \alpha^2} \begin{bmatrix} 1+\alpha^2 & \alpha \\ \alpha & 1+\alpha^2 \end{bmatrix} \begin{bmatrix} -\alpha \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} h[1] \\ h[2] \end{bmatrix} = \frac{1}{(1+\alpha^2)^2 - \alpha^2} \begin{bmatrix} -\alpha(1+\alpha^2) \\ -\alpha^2 \end{bmatrix} \rightarrow h[1] = -\frac{\alpha(1+\alpha^2)}{(1+\alpha^2)^2 - \alpha^2}, \quad h[2] = -\frac{\alpha^2}{(1+\alpha^2)^2 - \alpha^2}$$

$$\begin{aligned} \sigma_\varepsilon^2 &= E\{\varepsilon^2[n]\} = E\{(S[n] - \hat{S}[n])^2\} = E\{(S[n] - h[1]S[n-1] - h[2]S[n-2])^2\} \\ &= R_s[0] - h[1]R_s[1] - h[2]R_s[2] = \sigma_i^2 (1 + \alpha^2 + \alpha h[1]) \\ &= \sigma_i^2 \left(1 + \alpha^2 - \frac{\alpha^2(1+\alpha^2)}{(1+\alpha^2)^2 - \alpha^2} \right) = \sigma_i^2 (1 + \alpha^2) \left(1 - \frac{\alpha^2}{(1+\alpha^2)^2 - \alpha^2} \right) = \sigma_i^2 \frac{(1+\alpha^2)(1+\alpha^4)}{(1+\alpha^2)^2 - \alpha^2} \end{aligned}$$

When $\alpha = 0.8$ and $\sigma_i^2 = 2$:

$$\begin{bmatrix} 3.28 & -1.6 \\ -1.6 & 3.28 \end{bmatrix} \begin{bmatrix} h[1] \\ h[2] \end{bmatrix} = \begin{bmatrix} -1.6 \\ 0 \end{bmatrix} \rightarrow \begin{bmatrix} h[1] \\ h[2] \end{bmatrix} = \begin{bmatrix} -0.6401 \\ -0.3123 \end{bmatrix}$$

$$\sigma_\varepsilon^2 = R_s[0] - h[1]R_s[1] - h[2]R_s[2] = 3.28 + 0.6401 \cdot (-1.6) + 0.3123 \cdot 0 = 2.2558$$

(3) $\hat{S}[n] = h[1]X[n-1] + h[2]X[n-2] + b$

Since all the processes are zero mean we have that $E\{\varepsilon[n]\} = 0 \Rightarrow b = 0$.



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$$E\{\varepsilon[n]X[n-m]\} = 0, \quad m = 1, 2$$

$$E\{(S[n] - \hat{S}[n])X[n-m]\} = 0, \quad m = 1, 2$$

$$E\{(S[n] - h[1]X[n-1] - h[2]X[n-2])X[n-m]\} = 0, \quad m = 1, 2$$

$$R_{sx}[-m] - h[1]R_x[m-1] - h[2]R_x[m-2] = 0, \quad m = 1, 2$$

$$\begin{cases} h[1]R_x[0] + h[2]R_x[-1] = R_{sx}[-1] = R_s[1] \\ h[1]R_x[1] + h[2]R_x[0] = R_{sx}[-2] = R_s[2] \end{cases}$$

$$\begin{bmatrix} R_x[0] & R_x[-1] \\ R_x[1] & R_x[0] \end{bmatrix} \begin{bmatrix} h[1] \\ h[2] \end{bmatrix} = \begin{bmatrix} R_s[1] \\ R_s[2] \end{bmatrix}$$

$$R_x[m] = R_s[m] + R_w[m] = R_s[m] + \sigma_w^2 \delta[m] = R_s[m] + \delta[m]$$

$$\begin{bmatrix} 4.28 & -1.6 \\ -1.6 & 4.28 \end{bmatrix} \begin{bmatrix} h[1] \\ h[2] \end{bmatrix} = \begin{bmatrix} -1.6 \\ 0 \end{bmatrix} \rightarrow \begin{bmatrix} h[1] \\ h[2] \end{bmatrix} = \begin{bmatrix} -0.4346 \\ -0.1625 \end{bmatrix}$$

$$\begin{aligned} \sigma_\varepsilon^2 &= E\{\varepsilon^2[n]\} = E\{(S[n] - \hat{S}[n])S[n]\} = E\{(S[n] - h[1]X[n-1] - h[2]X[n-2])S[n]\} \\ &= R_s[0] - h[1]R_s[1] - h[2]R_s[2] = 3.28 + 0.4346 \cdot (-1.6) + 0.1625 \cdot 0 = 2.5846 \end{aligned}$$

which is obviously greater than the MSE in the previous case, because of the presence of additive noise $W[n]$.